

Test Data & Edge Report

Crypto swing engine - lab backtests, null tests, AI-gate verdict. 8 Oct 2026.

Everything below is the measured output of the lab backtests run on the ThinkPad (8 Oct 2026) over the full 4-year Binance history. No AI involved unless labelled. Costs are Binance taker fees + funding + slippage, netted into every R figure.

1. What was tested

- Data: 4 years of BTC/ETH/SOL/XRP/DOGE/LINK/BNB + alts, 1h bars aggregated to 1h/4h/8h/1d.
- Strategies: 8 specs (vol_breakout, keltner, ichimoku, trend_pullback, range_fade, ...) x 4 timeframes.
- Wallet sim: Rs5,000 start, sweep Rs3,000 each time equity hits Rs10,000, live risk settings.
- AI gate: 150/150 completed trades scored by the local LLM (Ollama) vs a no-AI baseline.
- Trial ledger: audit/trial_ledger.jsonl holds 2,879 configs (2,823 from search).

2. Edge reality (in-sample, net of costs)

Metric	Value	Reading
Book expectancy	+0.357 R / trade	real, positive
Cluster t-stat	6.5	well above the ~2 bar
95% CI on expectancy	+0.25 ... +0.47 R	excludes zero
Win rate	48%	trend-style: <50% but winners bigger
Trades in sample	3,430	statistically meaningful
Specs beating random	8 / 8	null_p = 0.00498
Best 4h spec (CI-low)	vol_breakout 0.195	most robust
Best 8h spec (CI-low)	keltner 0.187	best on 8h

3. What the sweep found

- Timeframes: edge is real on 4h and 8h. Ichimoku is net-negative below 4h.
- Exits: current exit (3x ATR stop / 3R target / 7-day time stop) is the best of the set tested.
- Cost stress: edge survives at +0.01 R extra cost per trade (still positive).
- Regimes: edge VANISHES in wild/high-vol markets (-0.32 R). The volatility filter is the key rule.
- Filters REJECTED: KAMA, McGinley, dead-market filter all hurt or added nothing.

4. AI gate verdict (150/150 done)

Metric	Value	Reading
AUC	0.499	coin flip (0.5 = no skill)
Veto precision	50.0%	vetoed are random
Take / Skip	36 / 114	skipped most trades
Lift 95% CI	-0.36 ... +0.81	includes zero -> no effect
Kept share of total R	32%	threw away 68% of profit

VERDICT: the AI gate is statistically worthless on this data. Do NOT wire it to live trades.

5. Honest limits

- All of the above is IN-SAMPLE. The forward test (cycle 2, started 7 Oct) has ~0 closed trades so far.
- Survivorship bias: only coins that survived are in the universe.
- Constants baked into the live path (STRATEGY_R, COIN_R, VOL_RANK_MAX=0.67) are in-sample fits.
- execution/ (453 lines) has never placed a real order - fake-exchange tested only.